6	(a)	(i)	greater deflection	M0	
			greater electric field / force on α-particle	A1	[1]
		(ii)	greater deflection	M0	
			greater electric field / force on α-particle	A1	[1]
	(b)	(i)	*either* deflections in opposite directions	M1	
			because oppositely charged	A1	
			or β less deflection	(M1)	
			β has smaller charge	(A1)	[2]
		(ii)	α smaller deflection	M1	
			because larger mass	A1	[2]
		(iii)	β less deflection because higher speed	B1	[1]
	(c)		*either* $F = ma$ and $F = Eq$ *or* $a = Eq / m$	C1	
			ratio = *either* $\dfrac{(2 \times 1.6 \times 10^{-19}) \times (9.11 \times 10^{-31})}{(1.6 \times 10^{-19}) \times 4 \times (1.67 \times 10^{-27})}$		
			or $[2e \times 1 / 2000\ \text{u}] / [e \times 4\text{u}]$	C1	
			ratio = 1 /4000 *or* 2.5×10^{-4} *or* 2.7×10^{-4}	A1	[3]